

# Problem Set # ORIGAMI

UMass Recreational Math Club

2025

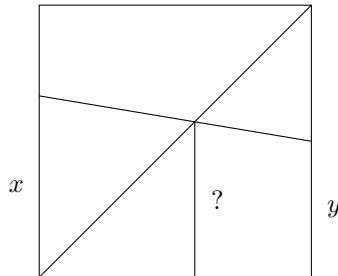
## 0.1 Constructing the rationals:

**Problem 0.** Construct  $3/4$ .

**Problem 1.** Find a construction for  $a/2^m$  for any integer  $a < 2^m$ . What is the minimum possible number of folds?

**Problem 2.** Can you construct  $a/b$  for any integers  $a, b$ ?

*Hint:*

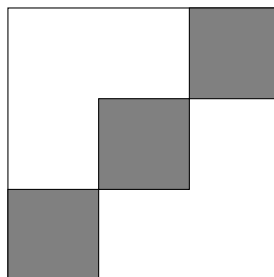


## 0.2 Folding Puzzles:

**Problem 3.** Given 4 collinear points, each an arbitrary distance from the next, construct a series of folds so that the points lie on the vertices of a rectangle.

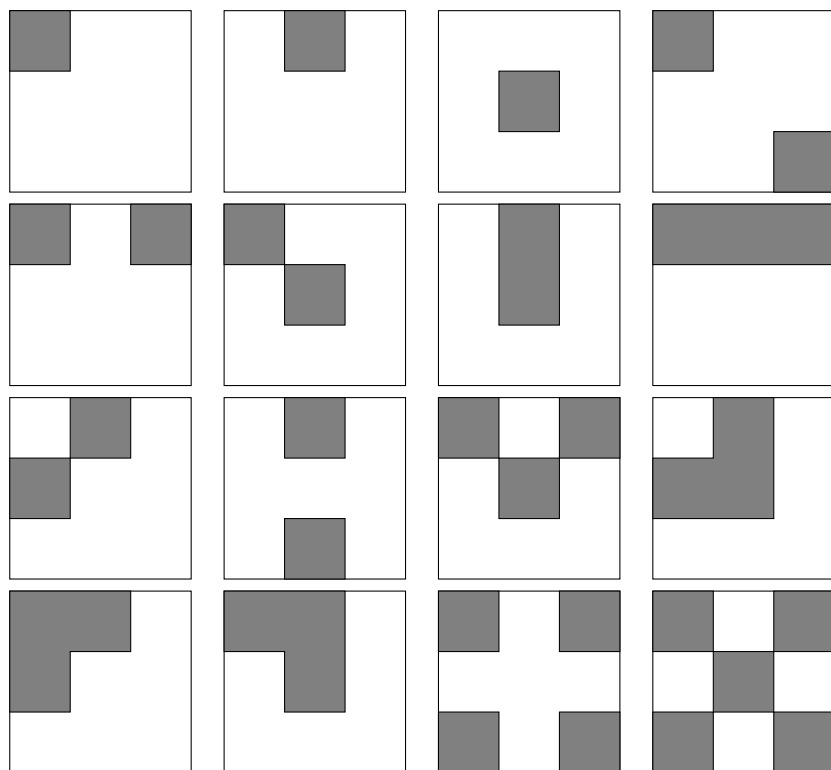
**Problem 4.** Starting with a square sheet of paper, fold it to produce a square having three-fourths its area. Only five folds are allowed.

**Problem 5.** Fold a square piece of paper, colored differently on each side, so that the result displays the color pattern shown below:



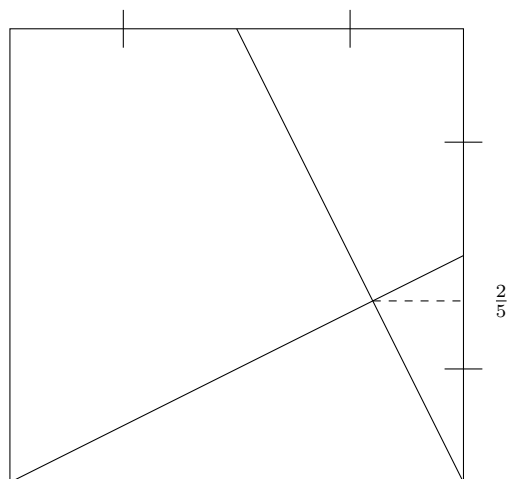
*Hint:* Divide your square paper into fifths. Once you've created the design, your final square should have an area that is  $9/25$  the area of the initial square.

**Problem 6.** Try to construct the following designs:



### 0.3 Extra hints!

#### Constructing fifths:



#### Miscellaneous hints:

- When you're sure they're right, press creases down with the back of the nail of your thumb.
- When duplicating a length (such as  $1/5$ ) several times, don't fold them sequentially like an accordion—rather, fold large pieces first (for example, halve  $4/5$  rather than duplicating  $1/5$ ).
- If you want to create a grid of lines, for example, don't repeat the construction in both dimensions—rather, fold a diagonal crease, then fold lines along the intersection points with the diagonal.